

Emergency Study Guide

Interior Construction & Detailing (Ballast, 6th ed.) · Ch 1 Partitions · Ch 2 Ceilings · Ch 5 Glazing · Ch 16 Lighting (16-11 to 16-21)

How to use this: Read this page first (the highest-yield facts). Then skim each chapter's **Key Terms** and **Main Points**. The tables are there for number-heavy questions. Page references (e.g., p 1-12) point back to the book.

Top Must-Knows (if you only read one page)

Partitions (Ch 1)

- **Gypsum wallboard** is the most common partition and ceiling material. The finished assembly is framing + board + trim + taped joints.
- **Type X** is fire-rated board. Gypsum resists fire because its **water of crystallization** turns to steam and absorbs heat.
- Most common commercial board is **5/8 in** on metal studs at **24 in o.c.** Use **1/2 in** on studs at 16 in o.c.
- Most common metal stud is **25 gage**. A higher gage number means thinner steel. Most common stud depth is **2½ in**, followed by 3½ in.
- **1-hr partition** (the most common rated wall) needs Type X, runs to the structure above, has all joints sealed, a **slip joint** at the top, and **fire dampers** in ducts.
- **2-hr partitions** separate occupancies and enclose stairs and elevator shafts.
- **STC** (Sound Transmission Class): higher is better. At STC 25 normal speech is heard clearly. At STC 50 loud speech is not audible.
- **Levels of finish run 0–5**. Level 1 is for plenums. Level 4 is the typical finish. **Level 5** adds a skim coat, for gloss paint or harsh lighting.
- **Resilient channel** separates the board from the studs to cut sound transmission.
- Three-coat plaster goes **scratch** → **brown** → **finish**. **Veneer plaster** is a thin coat over "blueboard" gypsum lath.
- **Demountable/movable partitions** have a higher first cost but a lower life-cycle cost. They carry **no fire rating** and can't hide plumbing.
- **Glass block** is never loadbearing, is laid in **stack bond**, needs expansion joints at the top and sides, and its supports may deflect no more than **L/600**.

Ceilings (Ch 2)

- **Suspended acoustical (lay-in)** is the most common commercial ceiling. It is cheap, fast to install, absorbs sound, and gives access to the plenum. Tiles absorb sound but don't block it.
- Grid: **main runners at 4 ft o.c.** on hanger wires, with **cross tees** every 2 ft. A **ceiling angle** trims the perimeter.
- Common tile sizes are **24×24** and **24×48 in**. **20×60** tiles fit a 5-ft planning module.
- The **plenum** is the space above the ceiling. If it is a return-air plenum, only noncombustible materials are allowed, and wiring must be in conduit or plenum-rated.
- **Seismic Design Category A–F** is set by a **structural engineer** from soil class, risk category, and location. **D/E/F** are the strictest.
- **Risk Category IV** covers essential facilities (hospitals with surgery or ER, fire and police stations). Its importance factor is 1.5.

Glazing (Ch 5)

- **Float glass** is the most common, basic (annealed) glass. **Tempered** glass is about **4× stronger**, breaks into small cubes, and **can't be cut after tempering**.
- **Laminated** glass uses a **PVB** interlayer that holds broken pieces together. It counts as safety glass, can be cut, and helps with acoustics and security.
- **Wired glass** is fire-rated but **not** safety glazing, because it can't be tempered.
- Five fire-rated glazing types: **wired, clear ceramic, gel-filled, tempered, and some glass block**.
- Fire-rated glazing label letters: **W** = wall, **OH** = openings, **D** = doors. **H/NH** = passes or fails the hose stream test. **T/NT** = passes or fails the temperature-rise test.
- **Safety glazing** is required in **hazardous locations**: doors, tubs and showers, glass within 24 in of a door, and railings. It is tested to **16 CFR 1201**. **Category II** (400 ft-lbf) is required for panels over 9 ft² and for all doors.
- **¼ in** is the standard interior glass thickness. Two-way mirrors need the observed room **5×** brighter than the observation room.

Lighting (Ch 16)

- Lamp families: **incandescent/halogen, fluorescent** (needs a **ballast**), **HID** (mercury vapor, metal halide, HPS), **LED**, and **neon/cold-cathode**.
- **Incandescent** is cheap, dims easily, and has warm color, but it has low efficacy, a short life, and high heat.
- **LED** lasts more than 50,000 hr, uses little power, and is digitally controllable.
- **HPS** is very efficient (80–140 lm/W) but gives yellow light.
- **Emergency lighting** must provide at least **1 fc** at the floor, with exit signs at every exit door.
- **ASHRAE 90.1** requires automatic shutoff in buildings over 5000 ft². Occupant sensors must turn lights off within **30 min**.
- **Daylighting** reaches about **2.5 × window head height** into a room. **Light shelves** bounce light deeper. Photocell dimming saves **50–60%** of perimeter lighting energy.
- Plumbing: **traps** block sewer gas and **vents** keep traps from siphoning. Drain lines slope at least **¼ in/ft**.

Chapter 1 — Partitions

Gypsum wallboard · Lath & plaster · Lime plaster · GRG · Masonry · Glass block · Demountable/movable

Key Terms

Partition — An interior wall that divides space. It also supports doors and glazing and houses electrical and plumbing.

Gypsum wallboard ("gyp board," drywall) — A gypsum core between paper facings. It is the most common, most versatile partition material.

Mastic application — Gluing board directly to a substrate such as block or concrete instead of fastening it to framing.

Tapered edge — The standard long edge of a board. It is recessed so tape and compound sit flush.

Square edge — An edge used for base layers, under veneer plaster, or where looks don't matter.

Shaft wall (core) board — A 1 in thick, 2 ft wide board for enclosing shafts.

Type X (fire-rated) board — Board with mineral additives (vermiculite, glass fiber) in the core that improve fire resistance. It is required in rated walls.

Water of crystallization — Water bound in gypsum (about 50% by volume). In a fire it turns to steam and absorbs heat.

Water-resistant board — A tile backer for damp areas. It is *not* allowed in saunas, steam rooms, gang showers, on ceilings, or over vapor retarders.

Abuse-resistant board — Board that resists surface damage, penetration, and security breaches. Used in lobbies, hospital corridors, and gyms.

Sound-deadening board — Board that reduces sound. One type has a viscoelastic polymer core between gypsum layers.

Mold-resistant board — Board with a paper face replaced by an inorganic (fiberglass mat) facing, so mold has nothing to feed on.

Foil-backed board — Board with a foil vapor barrier, used when remodeling exterior walls and ceilings. The foil faces the framing.

Prefinished board — Board with vinyl already applied, used in demountable systems.

Dry / wet bending — Bending board onto closely spaced studs for curves. Wetting the board allows a tighter radius.

Gage — Steel stud thickness. **A higher number means thinner steel**; 25 ga is the most common.

Runners (tracks) — C-shaped floor and ceiling channels that hold the studs.

Dimpled stud — A stud whose rolled dimples work-harden the steel. It uses less steel, and its limiting heights are 15–25% higher.

Slip joint — A top-of-wall joint that lets the structure above deflect without cracking the board. It is required for rated partitions.

Top deflection track — A proprietary top runner with intumescent material that allows up to 1 in of deflection.

Intumescent — A material that swells and chars in heat, sealing against fire and smoke.

H-stud — A stud for **area separation walls**, which act as firebreaks between apartments or condos.

Shaft wall liner / shaft lining — A 2-hr enclosure around stairs, elevators, or dumbwaiters, with board on one side.

L/120, L/240, L/360 — Deflection limits (L = length). L/120 is the code maximum. L/240 is recommended for brittle finishes.

Furring — Small framing members that give board something to attach to over other construction.

Hat channel — The most common metal furring, named for its hat-shaped profile.

Resilient channel — Furring with only one leg attached, which isolates the board and reduces sound transmission.

Z-furring — Z-shaped furring that gives more depth, for insulation or conduit.

Corner bead — L-shaped trim for all outside corners, covered with compound.

LC bead — Edge trim that goes on *before* the board and is finished with compound. Contractors also call it J or U bead.

L bead — Like LC bead but with no back flange, so it goes on *after* the board. Good for remodels.

U bead — Edge trim with no finishing. The metal stays visible, so it is the cheap option.

LK bead — Trim with a V edge that fits a kerf in adjacent wood trim.

Control joint — A joint in large expanses of board where movement is expected.

Reinforcing tape — Paper or fiberglass tape bedded in joint compound.

Textured coating — A finish such as "orange peel" that hides minor flaws.

Fire rating (1-, 2-, 3-, 4-hr) — How long an assembly stops fire, as measured by a lab test.

STC — Sound Transmission Class, a single-number rating of how well an assembly blocks sound. **Higher is better.**

Nonrated partition — A wall that runs from the floor to the suspended ceiling only. Common in commercial work.

Power-actuated fastener — A fastener driven by an explosive charge, used for the bottom runner on concrete.

Fire damper — A device in a duct that closes where the duct passes through a rated wall.

Chase wall — Two braced rows of studs spaced apart to hold plumbing or recessed items.

Soffit — A lowered section of ceiling, for example over cabinets, framed down from the ceiling.

High-attenuation partition — A double stud row that separates the two sides, reaching about STC 56–60.

Blocking — $\frac{3}{4}$ in plywood or wood between studs that supports cabinets and wall-hung items.

Kickers — Braces from the top of a partition up to the structure. Used for heavy doors and cabinets.

Synthetic gypsum — A byproduct gypsum, mostly from **flue gas desulfurization** at power plants. It makes up about 28% of the supply.

Plaster — A cementitious material applied wet that sets hard. The binder is gypsum (typical interiors) or portland cement (wet areas).

Stucco — Exterior portland cement plaster.

Keene's cement — Dehydrated gypsum mixed with lime putty, used for extreme abrasion resistance.

Scagliola — Plaster finished to look like marble.

Sgraffito — Layered colored plaster with the top layer scratched away to reveal the colors below.

Marezzo — Imitation marble made from colored Keene's cement.

Fresco — Painting on wet plaster before it sets.

Veneer plaster — A very thin coat ($1/16$ – $1/8$ in) over gypsum lath. It is fast, hard, and cheap, but not allowed in wet areas.

Gypsum lath ("blueboard") — Blue-faced board made to bond with veneer plaster.

Diamond mesh / flat rib / rib lath — Expanded metal laths. Diamond mesh bends into curves, flat rib is stiffer (for ceilings), and rib lath suits studless walls.

Scratch / brown / finish coat — The three plaster coats: scratch ($\sim 1/2$ in, bond), brown ($\sim 1/4$ in, leveling), and finish ($\sim 1/8$ in).

Lime (Venetian) plaster — Slaked lime putty plus crushed stone. It hardens by reabsorbing CO_2 , is flexible and mold-resistant, and has no VOCs.

Security partition — High-strength plaster over perforated steel sheet, used in jails and vaults.

GRG (GFRG/FRG) — Glass-fiber-reinforced gypsum cast in molds, used for column covers, domes, and moldings. It is decorative only, never structural.

Masonry — Brick, concrete block, glass block, clay tile, terra cotta, and gypsum block.

CMU (concrete block) — A hollow cement-and-aggregate block, typically $8 \times 8 \times 16$ in nominal ($7\frac{5}{8} \times 7\frac{5}{8} \times 15\frac{5}{8}$ in actual).

Nominal dimension — A size that includes the $\frac{3}{8}$ in mortar joint and is based on a 4 in module.

Structural clay tile — A glazed, hard, easy-to-clean unit used in hospitals and food plants.

Stack bond — Units laid with joints aligned vertically, as glass block is.

Glass block — A hollow or solid glass unit, nominally 4 in thick ("thin" is $3\frac{1}{8}$ in). It is never loadbearing.

L/600 — The deflection limit for members supporting glass block.

Demountable partition — Prefinished components that can be taken down and reused but otherwise act like a permanent wall.

Progressive / nonprogressive — In a progressive system, panels must be removed in order (cheaper). In a nonprogressive system, any panel can be removed.

Movable partition — A wall not attached to the structure, often part of a furniture system. The term is often used interchangeably with demountable.

COM — Customer's Own Material, a finish the client supplies.

Class A flame spread — The best (most restrictive) flame-spread class.

Main Points

Gypsum wallboard basics (p 1-1 to 1-5)

- Board widths are 4 ft and 4½ ft. Lengths are 8–14 ft, and installers use the longest practical length to reduce joints.
- Thicknesses are ¼, ⅜, ½, ⅝, and ¾ in.
 - ⅝ in: commercial work.
 - ½ in: residential work.
 - ⅜ in: double layers and remodels.
 - ¼ in: curves and resurfacing.
 - ¾ in: a single layer can give 2 hr.
- Board is usually hung **horizontally**, which means fewer joints, lower cost, and the board's strongest direction across the studs.
- Board can go on studs, on furring, glued with mastic, or on ceiling framing. Mastic needs clean, smooth, flat surfaces.
- Abuse-resistant board is made with core fibers, tough papers, Lexan or fiberglass backing, or veneer plaster. Impact ratings range from 60 to over 2100 ft-lbf.
- Commercial showers use tile on cement board or a mortar bed, not gypsum.

Framing (p 1-5 to 1-11)

- **Wood** (2×4, actually 1½×3½ in) is used in residential work, where walls can also carry loads. **Metal** is used in commercial work because it is noncombustible and light.
- Studs are spaced **16 or 24 in o.c.** 24 in is common for commercial nonloadbearing walls.
- Metal stud depths are 1⅝, 2½, 3⅝, 4, and 6 in. **2½ in** suits 8–10 ft ceilings. Use deeper studs for pipes, medicine cabinets, and pocket doors.
- Heavier gages (20, 22) are for tall walls, door openings, and cabinets. Structural studs are 12–18 ga.
- Assembly: fasten the runners first, then slip in the studs and screw or crimp them.
- Furring comes as 1×2 wood (minimum), hat channel, resilient channel, or Z-furring.
- **Six standard trim shapes:** corner bead, LC, L, U, LK, and control joint.
- Finishing sequence: tape bedded in compound, then more coats feathered wider, sanding between coats. Screw heads get the same treatment.

Fire & sound (p 1-12 to 1-27)

- Gypsum is naturally fire resistant, and its density makes it a good sound barrier.
- Where each fire rating is used:
 - **1-hr:** corridors, tenant separations, and house-to-garage walls.
 - **2-hr:** occupancy separations and stair or elevator shafts.
 - **3–4 hr:** some occupancy separations.

- Rules for 1-hr partitions:
 - Use Type X board and run the wall to the structure above.
 - Seal all joints and penetrations, and put fire dampers in ducts.
 - Allow only **one electrical box per stud space, on one side only**, with a maximum of 16 in².
- A 2-hr partition has two layers of Type X on each side, or a single layer of proprietary ¾ in board.
- Cracking is worst in **high-rises**, where buildings deflect and sway. Use slip joints at slabs and relief joints at mullions and columns (½ in gap).
- Ways to add sound control:
 - More layers of board.
 - Resilient channels.
 - Batt insulation.
 - A double stud row.
 - Sealing every penetration, since leaks ruin the rating.
- Coordinate: floor deflection, horizontal sway, heavy loads (blocking), partition depth, and anchoring (kickers).
- Code references: IBC Tables 721.1(2) and (3), the UL directory, and the *GA Fire Resistance Design Manual*.
- The IBC requires water-resistant board behind tub and shower tile and on water-closet compartment walls.

Plaster (p 1-28 to 1-33)

- Plaster's advantages over wallboard:
 - Single and double curves.
 - Ornate moldings.
 - Many textures.
 - Abrasion resistance.
 - Water resistance, with portland cement.
- Ingredients are a binder, lime, water, and aggregate (sand, perlite, vermiculite). Use a **portland cement** binder in wet areas, because moisture damages gypsum plaster.
- Three-coat plaster goes over metal lath. Two-coat plaster is mainly a base for tile.
- Veneer plaster is faster, cheaper, and harder, but not allowed in wet areas. Two layers of rated lath give 2 hr.
- Wet areas: use cement backer board, or full lath and plaster for very wet commercial spaces.
- Rated partitions need fire-rated gypsum lath.
- Lime plaster benefits:
 - Flexible, and small cracks self-heal.
 - Breathable.
 - High pH makes it a biocide that resists mold.
 - No VOCs, and it absorbs CO₂ as it sets.Over wallboard it needs a Level 4 or 5 finish.
- Plaster control joints go where plaster meets structure, in runs over **30 ft**, and at building expansion joints.
- Plaster walls need door frames of at least 16 ga steel.
- GRG tips:
 - Expose one side only.
 - Minimize the number of unique molds.
 - Use draft angles, not 90° corners.
 - Keep joints ≤ 3/8 in wide.
 - Extend curved pieces about 2 in into the flat board.

Masonry & glass block (p 1-33 to 1-37)

- Interior masonry walls are nonloadbearing. A structural engineer must confirm the floor can carry them. Size openings to the **4 in module**.
- CMU is laid in running bond with joint reinforcement every 16 in vertically. Filling the cells with grout or sand improves sound control.
- Glass block faces are 6×6, 8×8, 12×12, and 4×8 in. Thicknesses are 3 7/8 in (standard) and 3 1/8 in (thin, common indoors).
- Code requirements for glass block:
 - Units at least **3 in** thick.
 - Mortar joints 1/4–3/8 in.
 - Expansion joints allowing at least 3/8 in of movement.
 - Stiffeners when a panel exceeds the code maximum.
- Glass block usually has no fire rating. Some assemblies reach 45–90 min in openings up to 100 ft².
- Curved glass block walls need stiffeners at every change of curve direction.

Demountable / movable (p 1-36 to 1-41)

- Components: floor runner, ceiling runner, studs with clips, and prefinished vinyl gypsum panels (24 or 30 in wide, fitting 4 or 5 ft modules).
- **Operable** partitions (Ch 3) open and close often, like doors. **Demountable** partitions stay put.
- **Pros:**
 - Fast, clean reconfiguration with no wet compound.
 - Recycled and recyclable materials.
 - Single-source responsibility.
 - **Tax advantages** (treated as equipment).
 - Lower life-cycle cost.
 - Handles uneven floors.
- **Cons:**
 - **No fire rating.**
 - Can't hide plumbing.
 - Limited acoustics (under STC 50 as installed) and limited security.
 - Limited cabinet support.
 - Higher first cost.
- A spiked bottom track sits on carpet. The top track clips to the ceiling grid, so the space plan should align with the ceiling grid.
- Most systems are Class A flame spread.

Key Tables — Ch 1

Table 1.1 · Board thickness vs. framing spacing		
Use	16 in o.c.	24 in o.c.
Partitions / Ceilings	½ in*	⅝ in
Furring	½ in	½ in
Mastic (no framing)	⅝ in	⅝ in
*A fire rating may require ⅝ in Type X		

Table 1.5 · STC: what you hear through the wall	
STC	Effect
25	Normal speech heard clearly
30	Loud speech understood fairly well
35	Loud speech heard but not understood
42–45	Loud speech faint; normal speech not heard
46–50	Loud speech not audible

Table 1.4 · Max partition height (25 ga, L/120, most common)	
1⅝ in stud	8'-9"
2½ in stud	11'-3"
3⅝ in stud	13'-6"
4 in stud	14'-3"
6 in stud	15'-0"
20 ga studs go much higher (e.g., 3⅝ in reaches 19'-1")	

Levels of Gypsum Board Finish (GA-214)		
Lvl	Treatment	Use
0	Nothing	Temporary
1	Tape bedded in compound	Plenums, hidden areas
2	+1 coat	Tile base, garages
3	2 coats	Heavy texture, heavy wallcovering
4	3 coats	Typical: light texture, wallcovering, flat paint
5	Level 4 + skim coat over everything	Gloss or enamel paint, severe raking light

Table 1.6 · Common partitions (highlights)		
Build	Rating	STC
½ in board on 2×4 wood studs (residential)	¾ hr	~30
⅝ in board on 2½ in studs, to the ceiling only	none	~40
⅝ Type X on 2½ in studs to slab + insulation	1 hr	~47
2 layers ⅝ Type X each side	2 hr	~48
⅝ Type X, resilient channel, 3 in batt	1 hr	~54
Double wood studs on separate plates, 2 layers each side	2 hr	~56
Proprietary studs + 1 in liner panels	3 hr	~60

Table 1.9 · Min inside radius, curved glass block	
4 in block	32 in
6 in block	48½ in
8 in block	65 in
12 in block	98½ in

Max glass block panel (interior, IBC Table 1.8): Standard units: 250 ft², 20 ft high, 25 ft wide. Thin or solid units: 150 ft².

Bending radii: Wet bending allows tighter curves than dry. For ¼ in board, dry bending gives a 5 ft radius and wet bending gives 2 ft. Flexible ¼ in board bends even tighter.

Chapter 2 — Ceilings

Suspended acoustical · Seismic · Gypsum wallboard · Plaster · Linear metal · Decorative · Integrated · Abuse-resistant · Metal & wood

Key Terms

Plenum — The space above a suspended ceiling, used for ducts, wiring, and other services.

Return air plenum — A plenum that also returns air to the HVAC system. **Only noncombustible materials are allowed.**

Plenum-rated wiring — Wiring allowed to run exposed in a plenum. All other wiring must be in metal conduit.

Suspended acoustical ceiling — Mineral, glass-fiber, or wood panels in a metal grid hung on wires. It is the **most common** commercial ceiling.

Lay-in system — Panels resting on an exposed T-grid. The most common type.

Tegular — Lay-in panels with rabbeted (stepped) edges that drop below the grid.

Concealed grid — A grid hidden by the tiles, usually 1×1 ft tiles. Beveled edges hide misalignment.

Main runners — Primary grid members, hung on wires at **4 ft o.c.**

Cross tees — Secondary members that snap into the main runners every 2 ft.

Ceiling angle — L-shaped trim at the walls that supports edge tiles.

Concealed spline ceiling — Tiles with kerfed edges that slide onto a hidden grid.

Upward / downward accessible — Concealed-spline access methods. **Upward** (tiles pushed up onto Z-clips) is preferred, because the pull-down tool chips tiles.

Retention (hold-down) clips — Clips that keep tiles from lifting out. They are required for some rated ceilings.

Rated acoustical ceiling — Rated tiles on a rated grid with hold-down clips and expansion slots. It is only part of a rated floor/ceiling assembly.

Plenum sound barrier — Construction above a partition that stops sound from traveling over it through the plenum.

Seismic restraint — Bracing that keeps nonstructural items, such as ceilings, in place during earthquakes.

Seismic Design Category (A–F) — An IBC rating based on **soil class + risk category + location**. A is least restrictive and F is most. It is set by a structural engineer.

Risk (occupancy) category I–IV — How important it is for a building to survive an earthquake. IV covers essential facilities.

Importance factor — A multiplier per risk category: 1.00 for categories I and II, 1.50 for IV.

Soil class A–F — Soil ratings from A (hard rock) to F (special soil). **Assume D** when the soil is unknown.

CISCA — Ceilings & Interior Systems Construction Association, which publishes seismic ceiling guides (Zones 0–2 for Category C; Zones 3–4 for D/E/F).

ASCE 7 — The standard for minimum design loads, including seismic provisions.

Compression struts (posts) — Rigid vertical braces from the grid to the structure. Required in D/E/F.

Splay wires — Diagonal bracing wires at 45° that restrain lateral movement.

Spacer bars — Bars that keep grid ends from spreading at the perimeter.

Furring channel (hat) — Light channel that gypsum board ceilings screw to.

Suspended GWB ceiling — 1½ in steel channels at 4 ft plus furring, hung on wires, with board screwed on.

Ceiling radiation damper — A damper required where a diffuser penetrates a rated ceiling.

Access panel — A hinged steel door in a board ceiling (about 24 in square) for reaching valves and dampers.

Cloud — A separate panel suspended below the main ceiling.

Drapery pocket — A ceiling recess for drapery track or blinds.

Casing bead — Metal trim at plaster edges.

Perimeter relief (isolation) joint — A joint that separates a plaster ceiling from walls and structure.

Linear metal ceiling — Long prefabricated aluminum strips clipped to carriers. The gaps let air and sound through.

Decorative suspended ceiling — Baffles, wood strips, open grids, mirrored panels, and similar, hung on wires or rods.

Integrated ceiling — One system combining tile, lights, HVAC, sprinklers, and partition attachment, on a 4 or 5 ft module. Good where layouts change often.

Abuse-resistant ceiling — A ceiling built to resist impact, vandalism, frequent access, or entry. Used in schools, hospitals, and prisons.

Metal & wood suspended — Aluminum or wood-veneer panels (often perforated) on a grid.

FSC-certified — Wood from a Forest Stewardship Council–certified forest, which counts as sustainable.

VOCs — Volatile organic compounds that offgas from materials. Aluminum has none.

Main Points

General & suspended acoustical (p 2-1 to 2-7)

- Ceilings integrate lighting, partition attachment, air diffusers, and speakers, so their detailing is critical.
- **Three ceiling groups:**
 - Suspended from the structure.
 - Attached directly to the structure.
 - Exposed (the structure is the finish).
- **Residential:** board is screwed directly to the joists.
Commercial: wire-suspended systems.
- Advantages of suspended acoustical ceilings:
 - Low cost.
 - Fast installation.
 - Sound absorption (but **not** a sound barrier).
 - Flexibility.
 - Easy plenum access.
- Tile sizes: 24×24 and 24×48 are most common. 20×60 fits a **5 ft planning module** and is often used with demountable partitions.
- Suspended ceilings are installed level within $\frac{1}{8}$ in in 12 ft, which is more level than floors. That is why critical dimensions are measured from the ceiling down.
- Specify the accessible percentage for concealed-spline ceilings. For example, 25% means a quarter of the tiles can be removed.
- Concealed-spline joints show shadows under grazing light, so use ambient lighting.
- Code: tiles are almost all **Class A**. Installation follows ASTM **C635/C636** (E580 for seismic areas).
- A ceiling alone can't stop fire from spreading between floors.
- Plenum sound barrier options:
 - **Cheapest:** batt insulation on the tiles, 4 ft each side of the wall.
 - **Better:** run the partition to the structure, or hang a gypsum or lead barrier.

Seismic (p 2-6 to 2-13)

- About half of the U.S. has seismic risk. Highest-risk areas:
 - California.
 - Parts of Utah, Idaho, and Montana.
 - The **New Madrid** area (the Arkansas–Tennessee–Missouri–Kentucky junction).
 - The Carolina border.
 - Alaska and Hawaii.
- Risk categories:
 - **I:** low hazard (farm buildings, minor storage).
 - **II:** standard occupancies.
 - **III:** substantial hazard (assembly over 300 people, schools over 250, colleges over 500).

- **IV:** essential facilities (hospitals with surgery or ER, fire and police stations, shelters, air traffic control).
- **No special seismic design is needed for:**
 - Categories A and B.
 - Ceilings under 144 ft².
 - Gypsum board suspension systems.
 - Plaster and lath ceilings.
- In D/E/F, partitions must be braced **independently** of the ceiling. In C, they may attach to the grid only if the ceiling can move laterally.
- D/E/F grids attach on **two adjacent walls** only. Hanger wires go within 8 in of walls, and struts are placed 12 ft o.c.
- Proprietary seismic clips must be lab-tested.

Coordination (p 2-12)

- Coordinate recessed lights (clearance), ducts, sprinklers, speakers, smoke detectors, and drapery pockets.
- No combustibles in a return plenum, not even fire-retardant wood.
- **Don't paint acoustic tile**, because paint can kill its sound absorption. Clean or replace it instead.
- Walls that stop at the ceiling need diagonal bracing to the structure when they carry heavy loads.

Gypsum wallboard ceilings (p 2-13 to 2-17)

- Use ½ in board at 16 in o.c. and ⅝ in at 24 in o.c. Most commercial ceilings use 24 in spacing with ⅝ in board. Screws are preferred over nails.
- Commercial assembly: 1½ in channels at 4 ft o.c. on wires, with furring at 16 or 24 in.
- The whole suspension system must be **noncombustible**. Installation follows GA-216 and ASTM C840, C754, and C1007.
- In rated assemblies, diffusers need radiation dampers and access doors must be rated. In most commercial buildings the **floor structure**, not the ceiling, provides the rating.
- Board ceilings are smooth but hard to access, and access panels look bad, so limit them to areas without access needs.
- Board can form stepped, coffered, vaulted, and arched shapes.
- Combining board with acoustical tile gives the look of board plus the absorption and access of tile. Transitions use aluminum trim, bulkheads, or clouds.

Plaster ceilings (p 2-17 to 2-19)

- Used for wet areas (public showers, steam rooms) and ornate remodels. Plaster curves in two directions but costs more.
- Commercial assembly: 8 ga wires at 3 ft, 1½ in main runners at 4 ft, cross-furring, lath, and three coats.
- Moldings are either run in place with a template or cast and attached.
- Place relief joints at structure, penetrations, and dissimilar materials.
- **Control joints are required when the ceiling exceeds 50 ft (with relief joints) or 30 ft (without).**

Specialty ceilings (p 2-19 to 2-22)

- **Linear metal:** painted or anodized aluminum. The gaps allow sound absorption and return air.
- **Integrated:** higher first cost, but it saves money when layouts change often, so run a life-cycle cost analysis.
- **Abuse-resistant ceilings, by level:**
 - **Light** (commercial kitchens): faced tile.
 - **Heavy:** snap-in metal panels.

- **Maximum security:** heavy steel planks with tamper-proof fasteners.

Plaster and gypsum versions have **poor acoustics**.

Aluminum panel advantages:

- Light (1–2 lb/ft²).
- Can be custom shaped.
- Recyclable.
- **Class A.**
- Unaffected by moisture.
- **No VOCs.**
- Good acoustics when perforated and backed with batts.

Their drawback is a higher first cost.

Key Table — Seismic Requirements (Table 2.2)

Item	Category C	Categories D / E / F
Hanger wire	12 ga at 48 in o.c. (both)	
Main tee	Intermediate or heavy-duty	Heavy-duty
Grid-to-wall clearance	$\geq \frac{3}{8}$ in	$\geq \frac{3}{4}$ in
Perimeter ceiling angle width	$\geq \frac{7}{8}$ in	≥ 2 in
Grid attached on two adjacent walls	Not permitted	Required
Splay wires (4 at 90°, 45° slope) + compression struts at 12 ft o.c.	Not required	Required
Seismic separation joint	Not required	Areas > 2500 ft ²
Sprinkler head clearance	$\frac{3}{8}$ in all around	2 in opening or swing joint
Light fixtures mechanically attached	Required (both)	
Cable trays and conduit braced independently	Not required	Required

Chapter 5 — Glazing

Common glass types · Decorative glass · Special & fire-rated glazing · Framed · Frameless · Safety glazing · Coordination

Key Terms

Glazing — Installing glass in framing. The word also means the glass itself.

Float glass — The most common glass, made by floating molten glass on molten tin and then annealing it. It replaced plate glass in the 1960s.

Annealed — Slowly cooled. Annealed float glass is the baseline strength.

Heat-strengthened glass — About 2× as strong as annealed. Mostly used for exterior windows. *Not* safety glass.

Tempered glass — Glass heated to about 1200°F and quenched. It is about **4× stronger**, breaks into small cubes, and counts as **safety glass**. It can't be cut after tempering.

Tong marks — Small marks near an edge from tongs used during vertical tempering. Hide them in the frame, or specify horizontal tempering.

Bow — Warping from heat treatment. It is the most noticeable distortion in tempered glass.

Laminated glass — Glass layers bonded with a **PVB (polyvinyl butyral)** interlayer that holds pieces together when broken. It is safety glass, can be cut, and suits acoustic and security uses.

Polycarbonate — A tough thermoplastic laminated in for ballistic or impact resistance.

Wired glass — Glass with embedded wire mesh. It is fire-rated, but **not safety glass** because it can't be tempered.

Figured / patterned glass — Glass rolled with a pattern that obscures vision without fully blocking it.

Mirrored glass — Glass with a metallic film and a protective backing. Mount it mechanically, not with mastic alone.

Two-way (transparent) mirror — A mirror on one side and a window from the other. The observed room needs at least **5×** more light (10× is better), and the coating faces the brighter room.

Beveled glass — Glass with a sloped, polished decorative edge.

Etched glass — Glass with part of the surface removed by acid or sandblasting. Uniform etching is **frosted**, and deep etching is **carved**.

Stained glass — Glass colored with metal oxides and set in lead or zinc **comes** (H-shaped strips).

Hand-blown glass — Glass blown as a cylinder, then cut and flattened. It has unique irregularities.

Flashed / crackle / seedy — Hand-blown variations: flashed has a thin colored layer, crackle has a webbed pattern from dipping in water, and seedy has small bubbles.

Cathedral glass — Machine-rolled glass, textured on one side and transparent.

Opalescent glass — Translucent, marbled glass with multiple colors mixed in.

Painted glass — Vitreous paint fired onto glass. It allows finer detail than stained glass.

Dichroic glass — Glass with thin metal-oxide film layers. It reflects one color and transmits another, shifting with viewing angle.

Cast glass — Molten glass poured into a sand mold. It is thick and usually comes in small pieces.

Kiln-formed glass — Glass heated until it sags over a mold, giving relief on one side and a smooth other side.

Clear ceramic — A fire-rated clear ceramic, not actually glass. It is more impact-resistant than wired glass.

Gel-filled glass — Tempered layers with a gel between them that turns opaque and insulates in a fire. It is rated up to 2 hr and is expensive.

Fire-protective (tempered) glass — Rated for **30 min** max, because it fails the hose stream test. It is ¼ in thick and still meets the 16 CFR 1201 impact standard.

W / OH / D — IBC labels for where fire-rated glazing may go: **W**all, **O**pening (windows, sidelights), and **D**oor. The rating in minutes follows.

H / NH, T / NT — H or NH means the glazing meets or fails the **H**ose stream test. T or NT means it meets or fails the **T**emperature-rise test. Example: "D-H-NT-45."

Bent glass — Flat glass heated and shaped into curves. Maximum size is about 4×8 ft.

Decorative laminated glass — Laminated glass with a colored film, fabric, rice paper, or mesh interlayer.

Electrochromic glazing — Glass that tints clear to dark when current is applied. It never becomes fully opaque, so it can't provide privacy, but it saves energy.

SPD (suspended-particle device) — Glass that varies continuously from clear to fully opaque.

PDLC (polymer-dispersed liquid crystal) — Glass that switches from clear to cloudy white. It gives **privacy** but doesn't control daylight.

Photochromic / thermochromic — Glass that tints in response to light or to temperature.

Low-iron glass — Glass with no green tint, for ultra-clear uses and visible edges.

Antireflective glass — Glass that cuts reflection from about 8% to under 1%. Used for display cases and storefronts.

LED-embedded glazing — Laminated glass with LEDs inside, powered through a clear conductive coating.

Framed glazing — Glass in wood, steel, or aluminum frames. The most common interior type.

Setting blocks — Small blocks the glass sits on, placed at the quarter points of the sill.

Removable stop — The frame piece that holds the glass in. Put it on the less important side of the wall.

Hollow-metal (steel) frame — A 16–18 ga bent steel frame, needed for fire-rated openings. Its stops show screws.

Aluminum frame — An extruded frame with sharp corners and snap-in stops (no exposed fasteners). It has no fire rating.

Face clearance / edge clearance / bite — The gap between the glass face and the stop, the gap at the glass edge, and how far the glass sits inside the frame.

Frameless glazing — Glazing with head and sill frames recessed and jambs hidden or eliminated, so the glass appears to float.

Butt joint / T-joint — A butt joint is two glass edges sealed with silicone. A T-joint is used where three pieces meet; one piece extends past the others.

All-glass system — Glass walls supported only at the top and bottom, braced by **glass fins (mullions)**.

Patch fittings — Metal fittings that join glass pieces or doors without framing.

Shake — How much a glass panel flexes or vibrates. It affects the thickness choice.

Safety glazing — Tempered or laminated glass required in **hazardous locations**.

16 CFR 1201 — The federal safety glazing standard. Category I is tested at 150 ft-lbf and Category II at 400 ft-lbf.

ANSI Z97.1 — An alternate safety test (Classes A and B), **not allowed** for doors or tub, shower, or sauna enclosures.

Main Points

Common glass (p 5-1 to 5-3)

- Float glass comes clear or tinted (bronze, gray, green). Thicknesses run $\frac{1}{8}$ to $\frac{3}{4}$ in. Indoors it is used only where safety glass isn't required.
- Tempered glass comes $\frac{1}{8}$ to $\frac{7}{8}$ in thick, and **$\frac{1}{4}$ in is common indoors.**
- Tempered glass **can't be cut, drilled, or deeply etched afterward**, so order it to exact size.
- When butting large frameless tempered panes, orient the bows the same way.
- Tong marks sit within $\frac{1}{2}$ in of the edge for glass up to $\frac{3}{8}$ in thick, and within $\frac{3}{4}$ in for thicker glass.
- **Laminated glass:**
 - Can be float, tempered, or heat-strengthened glass.
 - Can be cut in the field.
 - Good for acoustics and security (bullet-resistant with polycarbonate).
 - Low impact resistance unless tempered glass is used.
 - Interior sizes run $\frac{9}{32}$ to $\frac{3}{4}$ in. Maximum sheet is 84×160 in.
- Standards:
 - **ANSI Z97.1** and **16 CFR 1201**: safety glazing.
 - **ASTM C1036**: flat glass.
 - **ASTM C1048**: tempered and heat-strengthened glass.
 - **NFPA 80**: fire doors and windows.
- Wired glass.
- Clear ceramic.
- Gel-filled glass.
- Tempered glass.
- Some glass block.
- Ratings by type:
 - **Wired glass** ($\frac{1}{4}$ in, steel frame): **45 min.**
 - **Clear ceramic** (only $\frac{3}{16}$ in thick): 1 hr up to 1296 in², and 3 hr up to 100 in².
 - **Gel-filled** (tested to ASTM E119): up to **2 hr**, and it blocks heat.
- A fire barrier must stop **flame**, stop **smoke**, and limit **heat transfer**. Most glass does only the first two, and radiant heat can still ignite things nearby.
- **1-hr corridor walls** need **45-min** glazing, and their doors may be **20-min**. Other 1-hr walls need **60-min** glazing.
- Test standards: W uses ASTM E119, OH uses NFPA 257, and D uses NFPA 252.
- Rated glazing needs rated framing tested with the glass, which is bulkier than standard framing.

Decorative glass (p 5-4 to 5-7)

- **Wired glass** meets NFPA 80 and is UL-labeled. Openings are limited to **1296 in² (9 ft²)** with a **54 in** maximum dimension. It is not safety glazing.
- Mount mirrors mechanically (J-moldings, clips, rosettes) with rubber sleeves so metal never touches the glass.
- Minimum glass thickness for etching:
 - Light etching: $\frac{1}{8}$ in.
 - Deeper etching: $\frac{1}{4}$ in.
 - Medium carving: $\frac{3}{8}$ in.
 - Heavy carving: $\frac{1}{2}$ in or more.

Deep etching must be done before tempering.

- **Lighting etched glass:** light the frosted part itself, from above or at the edge. Edge lighting fails beyond about **18 in.**
- Stained glass in ceilings or damage-prone spots should be protected with tempered or laminated glass.
- Kiln-formed glass is $\frac{1}{4}$ – $\frac{1}{2}$ in thick, up to 4×8 ft, and can be tempered.
- Dichroic glass: a single film stack gives no red or green, and two stacks give all colors.

Fire-rated glazing (p 5-6 to 5-9)

- **Five types:**

Special glazing (p 5-9 to 5-11)

- Three switchable types:
 - **Electrochromic:** tints but no privacy.
 - **SPD:** clear to fully opaque.
 - **PDLC:** privacy, but no light control.

All are very expensive. Electrochromic and SPD save 20–30% of energy.

- Low-iron glass is for absolute clarity. Antireflective glass is for displays.
- Exterior glass type affects **UV fading** of interior finishes, so specify fade-resistant materials.

Framed glazing (p 5-10 to 5-15)

- Interior glass has no wind load or weather sealing, so framing sizes are less critical.
- **Wood frames:**
 - At least ¼ in of shim space.
 - One stop removable.
 - Two setting blocks at the sill quarter points.
 - Silicone or neoprene glazing tape.
- Glass thickness:
 - ¼ in: the standard.
 - ⅝–3/16 in: small pass-throughs.
 - **Thicker:** large openings.
 - **Tempered or laminated:** hazardous locations.
- **Steel** frames are for fire ratings and matching hollow-metal doors. They have rounded corners.
- **Aluminum** frames have sharp corners, snap-in stops, and anodized finishes.

Frameless & all-glass (p 5-13 to 5-21)

- Full-height glazing 8–10 ft tall needs at least ¼ in **tempered**, and ⅝ in is preferred. Use laminated glass for sound control.
- Leave extra space at the head so the glass can be lifted in and swung upright.
- Silicone sealant comes clear or black. Clear can show bubbles, so black often looks better.
- Use a **T-joint** where three pieces meet, because sealant can't be installed well at a single point.
- All-glass systems reach **30 ft** when the weight rests on the base, or **75 ft** when hung from clamps.
- Blinds in a **recessed blind pocket** give conference rooms privacy.
- A partial-height wall doesn't need safety glass if the sill is ≥ **18 in** above the floor.

Safety glazing — hazardous locations (IBC 2406)

- Hazardous locations:
 - Swinging, sliding, and bifold doors.
 - Unframed doors.
 - **Tubs, showers, spas, saunas, steam rooms, and pools**, where the glass is < 60 in above the standing

surface and within 60 in of the water.

- **Guards and railings**, always.
- Panels near walking surfaces (the 18/36 in rules).
- Glass within **24 in of a door**.
- Which category applies:
 - Panels **over 9 ft²** need Category II.
 - Panels under 9 ft² need Category I, **except** doors, pool and sauna enclosures, and sliding patio doors, which need Category II.
 - **Every glass door must be tempered.**
- Canada requires tempered or laminated glass in doors, showers, and sidelights wider than 500 mm.

Coordination (p 5-21 to 5-22)

- Make sure large sheets fit through the building to reach the space.
- Specify horizontal tempering to avoid tong marks.
- Note which way patterned or etched glass faces.
- Match frameless glass thickness to adjacent glass doors.
- Add **markings** on full-height glass so people don't walk into it. Furniture or planters can also act as barriers.
- Put stops on the side of the wall where appearance matters least.
- Leave at least **1½ in** between exterior glass and blinds, with top and bottom gaps, so trapped heat can't crack the glass. Put HVAC outlets on the room side of the blinds.

Key Tables — Ch 5

Table 5.2 · Max fire-resistance-rated glazing			
Rating	Max area	Max size	Thick
45 min	4500 in²	95×95 in	¾ in
60 min	5605 in²	95×95 in	15/16 in
90 min	3724 in²	90×90 in	1-7/16 in
120 min	3724 in²	111×111 in	2⅛ in

Table 5.3 · Interior glass clearances (in)			
Glass	Face	Edge	Bite
⅛–¼	⅛	¼	⅜
5/16–⅜	3/16	5/16	7/16
½	¼	⅜	7/16
¾	¼	½	⅝

Table 5.4 · Butt-joint widths for frameless glass (min / max)				
⅜ in glass: ⅜ / 7/16	½ in: ⅜ / 7/16	⅝ in: ⅜ / ½	¾ in: ½ / ⅝	⅞ in: ½ / ⅝

Chapter 16 — Lighting (16-11 to 16-19) + Plumbing Start (16-20)

Mechanical & Electrical Systems Coordination · Conduit & communications (16-10) · Light sources · Lighting systems · Controls · Daylighting · Plumbing

Key Terms

EMT — Electrical metallic tubing, the most common steel conduit. It is light and bends easily.

IMC / RMC — Intermediate and rigid metal conduit, the heavier steel types.

NEC — National Electrical Code, which governs conduit sizing and bending.

Plenum-rated cable — Cable allowed without conduit in plenums. It costs more.

Incandescent lamp — A tungsten filament in a gas-filled bulb.

Tungsten-halogen (quartz) — An incandescent lamp with halogen gas. The halogen cycle redeposits tungsten on the filament, giving longer life, whiter light, and steady color.

Low-voltage lamp — A lamp that usually runs at 12 V and needs a **transformer**. It is compact and gives tight beam control.

MR lamp — A miniature reflector lamp (low-voltage) that allows small fixtures.

R / ER / PAR lamps — Reflector lamps. They need large clearances from combustibles.

Fluorescent lamp — An efficient linear source that needs a **ballast**.

T8 / T5 — Fluorescent tubes 1 in (T8) and 5/8 in (T5) in diameter. They replaced the T12. The number is the diameter in eighths of an inch.

EPAct 1992 — The Energy Policy Act, which ended production of many T12 lamps.

Compact fluorescent (CFL) — A bent T4 or T5 tube lasting about 10,000 hr. Some have a screw base to fit older fixtures.

GU24 — A two-pin base that can't accept incandescent lamps. Its lamps have CRI ≥ 80 , give 40–70 lm/W, and last 10–30k hr.

Ballast — A device that supplies starting and operating voltage to fluorescent and HID lamps. Leave room for it in details.

HID — High-intensity discharge lamps, which pass an arc through high-pressure gas: **mercury vapor**, **metal halide**, **HPS**.

Mercury vapor — An HID lamp with blue-green light, long life, and **poor color**.

Metal halide — An HID lamp with better color and efficacy but shorter life. It has the best mix of HID features, but its color shifts over its life.

Ceramic metal halide (CMH) — Metal halide with a ceramic arc tube, giving better color and consistency. Used in retail and high ceilings. It costs more and is hard to dim.

HPS — High-pressure sodium, with a **yellow** light. Very efficient (80–140 lm/W), lasting 10–24k hr.

LED — A light-emitting diode (solid-state). It lasts over 50,000 hr, uses little power, produces no heat in the beam, and can be digitally controlled.

OLED — An organic LED, part of the same solid-state family.

Neon — Gas-filled tubing bent into shapes. The gas sets the color, and it needs a transformer.

Cold-cathode — Like neon but larger (about 1 in diameter), more efficient, and available in whites.

Emergency lighting — Code-required lighting for evacuation during a power failure.

Troffer — A recessed fluorescent fixture in a ceiling grid.

Cove lighting — Hidden lamps in a cove that wash a ceiling or wall with light.

Three-way / four-way switch — Switches that control the same lights from two or three locations.

Multilevel (two-level) switching — Controls that reduce light in steps, for example half the lamps of a fixture on each switch.

Time-of-day controller — A control that turns lights off on a schedule, with an override.

Occupant sensor — A passive infrared or ultrasonic sensor that turns lights on and off based on occupancy.

Daylight compensation — Photocells that dim or switch lights as daylight changes.

ASHRAE/IES 90.1 — The energy standard for buildings except low-rise residential. It sets lighting control rules.

UL — Underwriters Laboratories. Custom fixtures built with UL-listed parts avoid extra approval.

Daylighting — Using natural light to supplement or replace electric light.

Clerestory — A high band of windows between two roof levels.

Effective daylight depth — About $2.5 \times h$, where h is the height from the floor to the window head.

Light shelf — A horizontal shelf above eye level that bounces daylight onto the ceiling and deeper into the room while shading the lower glass.

Trap — A water-filled fitting that keeps **sewer gas** out.

Vent — A pipe open to the outside that releases gas and equalizes pressure so traps aren't siphoned dry.

Soil stack / waste stack — Vertical drains. A soil stack carries toilet (human) waste, and a waste stack carries only other waste.

Stack vent / vent stack — A stack vent is the part of a stack above the highest fixture. A vent stack is a separate vent pipe in multistory buildings.

Fixture branch — A pipe from a trap to a stack.

Cleanout — An access fitting for clearing a drain.

Main Points

Power & communications (p 16-10 to 16-11)

- EMT is the most common conduit. The NEC sets conduit size by the number of wires, and a larger conduit needs a larger bend radius.
- Telephone and data outlets go on the same plan as power. The designer locates them, and the engineer or contractor circuits them.
- Communication wiring is low voltage, so it has fewer restrictions and may run without conduit.

Light sources — what to detail for (p 16-11 to 16-15)

- Detailing considerations: **size, weight, location, mounting method, and heat output.**
- Cove lights need continuous support. Recessed downlights need clearance above the ceiling.
- **Incandescent:**
 - **Pros:** cheap, compact, easy to dim, frequent switching doesn't shorten life, warm color, easy to control with reflectors and lenses.
 - **Cons:** low efficacy, short life, high heat.
- **Halogen:**
 - Whiter, more efficient light with steady color.
 - Runs hot and can **shatter**, so it must be enclosed by glass or a screen.
 - Available at 120 V or low voltage.
- **Low-voltage:** small, with precise beams, good for lighting objects from far away. The transformer adds bulk and cost.
- **Fluorescent:**
 - Efficient and linear.
 - Needs a ballast (leave room for it).
 - T8 and T5 replaced T12.
 - CFLs and GU24 lamps replace incandescent lamps.
- Energy Star residential fixtures can't use the Edison screw base.
- **HID** lamps light big spaces such as gyms, garages, and industrial areas. Metal halide is the best all-around HID, CMH gives the best color, and HPS is the most efficient but yellow.
- Metal halide bulbs come clear (for optical control) or coated/diffuse (for better color).
- **LED:** bright, long life, low energy, colors, and digital control, but higher cost. LED replacements exist for A, R, PAR, and MR lamps and for fluorescent tubes.
- **Neon and cold-cathode** both need transformers and small mounting brackets.

Emergency lighting (p 16-12 to 16-13)

- Governed by the IBC, NEC, *Life Safety Code*, and Canadian codes.
- Required in:

- Exit stairs and corridors.
- Assembly spaces.
- Educational and hazardous occupancies.
- Large occupant loads.
- Lighting and sign requirements:
 - **At least 1 fc (10.8 lux) at the floor.**
 - **Exit signs** at every exit door, with directional signs at turns.
 - Luminous egress path markings in some high-rises.
- Remodels must reconnect or relocate emergency circuits and exit signs.

Lighting systems (p 16-15)

- Recessed fixtures need plenum clearance. Troffers are usually fine, but large downlights may clash with ducts.
- Choose low-clearance fixtures, and check both the drawings and the actual ceiling space.
- **Cove lighting:**
 - Use fixtures made for coves.
 - Keep lamps **out of sight lines** from across the room.
 - Set lamps far enough from the surface to avoid hot spots.
 - Leave room for relamping.

Control devices (p 16-16 to 16-17)

- Switching depends on:
 - The space's function.
 - How much individual control is needed.
 - Switch location (usually at the entry door).
 - Energy conservation.
 - The load per circuit.
- Dimmed lights need their own circuit. Fluorescent dimming costs more.
- Commercial voltages: **incandescent runs on 120 V** and **fluorescent on 277 V**, which is more efficient.
- Dimmers alone don't save much energy, so **automatic controls are required**.
- Daylight compensation needs the perimeter zone to get at least half its light from daylight for several hours a day. It can save up to 60%.
- **90.1 automatic shutoff** applies to buildings **over 5000 ft²**, using one of:
 - Time-of-day controls ($\leq 25,000$ ft² and one floor per control).
 - Occupant sensors that turn lights off within **30 min**.
 - A signal from another system.
- Exceptions: 24-hour operations, patient care, and places where shutoff would endanger safety or security.
- Per-space controls under 90.1:
 - Every enclosed space needs at least one control.
 - **Classrooms, conference rooms, and break rooms need occupant sensors.**
 - Each control covers at most 2500 ft² (spaces $\leq 10,000$ ft²) or 10,000 ft² (larger spaces).
 - The override lasts up to **4 hr**.
- Separate controls are required for:
 - Display and accent lighting.
 - Hotel rooms (a master switch at the entry).
 - Task and under-cabinet lighting.
 - Nonvisual lighting (food warming).
 - Lighting for sale or demonstration.
- Exterior lighting needs a photosensor or astronomical time switch.

- Codes limit the watts per circuit, so large, bright spaces need several circuits.

Custom lighting & daylighting (p 16-18 to 16-19)

- For custom fixtures, the designer sets the intent and a lighting designer details it. Use **UL-listed parts**, because fully custom fixtures need costly testing.
- Lighting plus the cooling it requires uses **30–40%** (up to 50%) of a commercial building's energy.
- Daylighting saves energy and benefits occupants. Photocells cut perimeter lighting energy by 50–60%.
- **Designer strategies:**
 - Open plans with window areas kept open.
 - Low partitions and interior glass.
 - Window coverings that let daylight in.
 - Reflective surfaces.
 - Electric lighting coordinated with daylight.
- **Variables:**
 - Facade orientation.
 - Sky conditions (sun angle, clouds, time of day).
 - Glass area, head height, and transmittance.
 - Room and ground reflectance.
 - Obstructions such as overhangs and trees.
- **Problems:**
 - Heat gain or loss.
 - **Glare**, which is worst on south sides.
 - Unbalanced light.
 - No energy saved unless the electric lights actually turn off.
- Daylight drops off beyond **2.5h**, so put enclosed rooms deeper in the floor plate.
- **Light shelves** work best when they sit partly inside and partly outside the glass. The designer can add interior shelves in older buildings with tall windows.

Plumbing (p 16-20 to 16-21)

- Plumbing has two parts: **supply** (pressurized, small pipes, easy to route) and **drainage** (gravity-driven, needs slope and vents, harder to route).
- Traps are omitted only when a fixture has a built-in trap (toilets) or adjacent fixtures share one (a double sink).
- Place fixtures **as close as possible to existing stacks**.
- Drain slope is at least **¼ in/ft**, or **⅛ in/ft** for pipes over 3 in. The ceiling depth therefore limits how far a fixture can be from the stack.
- Codes limit the trap-to-vent distance (IPC Table 16.2). Island sinks need special venting.

Numbers to Memorize — Quick Cheat Sheet

Partitions		Ceilings	
Common commercial board	5/8 in on studs at 24 in o.c.	Main runners / hanger wires	4 ft o.c.
Residential board	1/2 in on studs at 16 in o.c.	Cross tees	Every 2 ft
Single-layer 2-hr board	3/4 in	Lay-in tiles	24×24, 24×48; 20×60 for 5 ft module
Board widths	4 ft, 4 1/2 ft	Concealed-grid tiles	12×12 in
Common stud gage	25 ga (higher number = thinner)	Level tolerance	1/8 in in 12 ft
Common stud depths	2 1/2 in, then 3 5/8 in	Seismic categories	A–F; A and B need nothing special
Wood stud actual size	1 1/2 × 3 1/2 in	Unknown soil	Assume Class D
Deflection limits	L/120 code max; L/240 for brittle finishes	Importance factor, Cat IV	1.5
Electrical box in rated wall	≤ 16 in ² , one side per stud space	Seismic exemption	Ceilings under 144 ft ²
Relief joint gap	1/2 in	Seismic hanger wire	12 ga at 48 in o.c.
Slip joint	≤ 1/2 in gap; deflection track allows up to 1 in	D/E/F wall clearance / angle	3/4 in / 2 in
Acoustic batt beyond partition	4 ft each side	C wall clearance / angle	3/8 in / 7/8 in
Plaster coats	Scratch 1/2 in, brown 1/4 in, finish 1/8 in	D/E/F struts	12 ft o.c., starting 6 ft from walls
Veneer plaster	1/16–1/8 in	First hanger from wall	≤ 8 in
Plaster control joints	Runs over 30 ft	D/E/F separation joint	Areas over 2500 ft ²
CMU actual size	7 5/8 × 7 5/8 × 15 5/8 in	Plaster ceiling control joint	50 ft with relief joints, 30 ft without
Masonry module	4 in	Access panel	~24 in square
Glass block min thickness / joints	3 in / 1/4–3/8 in	Integrated ceiling module	4 or 5 ft
Glass block support deflection	L/600		
Glass block max panel	250 ft ² (thin: 150 ft ²)		
Demountable panel widths	24 or 30 in (4 or 5 ft modules)		
Synthetic gypsum share	~28%		

Glazing	
Tempered vs. float strength	~4×; heat-strengthened ~2×
Tempering temperature	~1200°F
Standard interior glass	¼ in
Full-height frameless	¼ in minimum, ⅜ in preferred
Wired glass opening limit	1296 in² (9 ft²), max 54 in; 45 min
Clear ceramic	1 hr ≤ 1296 in²; 3 hr ≤ 100 in²
Gel-filled	Up to 2 hr
Tempered fire-protective	30 min max
1-hr corridor wall glazing / door	45 min / 20 min
Other 1-hr walls	60 min
Safety Category I / II	150 / 400 ft-lbf
Category II required	Panels over 9 ft², and all doors
Hazardous: near door	Within 24 in
Hazardous: tub or shower	Under 60 in high and within 60 in
Partial-wall sill exemption	≥ 18 in above floor
Two-way mirror light ratio	5× (10× better)
Edge-lit etched glass limit	~18 in
All-glass height	30 ft (base-supported) / 75 ft (hung)
Wood frame shim space	≥ ¼ in
Blind-to-glass gap	≥ 1½ in
Antireflective glass	8% reflection → under 1%
Switchable glass energy savings	20–30%

Lighting & Plumbing	
Emergency lighting	≥ 1 fc (10.8 lux) at floor
LED life	Over 50,000 hr
CFL life	~10,000 hr
GU24	CRI ≥ 80, 40–70 lm/W, 10–30k hr
HPS efficacy / life	80–140 lm/W / 10–24k hr
T8 / T5 diameter	1 in / ⅝ in
Low-voltage lamps	Usually 12 V, need transformer
Commercial voltages	120 V incandescent / 277 V fluorescent
90.1 auto-shutoff threshold	Buildings over 5000 ft²
Occupant sensor shutoff	Within 30 min
Time-of-day control area	≤ 25,000 ft², one floor
Area per control	2500 ft² (spaces ≤ 10k ft²) / 10,000 ft²
Override duration	Up to 4 hr
Lighting share of building energy	30–40% (up to 50%)
Daylight savings (perimeter)	50–60%
Daylight depth	~2.5 × window head height
Drain slope	¼ in/ft (⅛ in/ft if over 3 in)

Heads-up: These notes were made from phone photos. A few numbers were hard to read and are worth checking against the book if there's time:

- Wallboard control-joint spacing (p 1-11).
- One row of Table 1.2 that reads "5 ft (3050 mm)."
- Structural clay tile thicknesses (p 1-34).

Everything else was clearly legible.